

Data Analysis

Task 2: Regression

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Task description

- ▶ Dataset
 - ▶ Pick the dataset you have previously selected for the 1st task.
 - ▶ If, for some reason, the dataset is unsuitable for this task - you can select another (but confirm it with Tomas)
- ▶ DIY regression:
 - ▶ Implement linear regression from scratch by using **gradient descent** algorithm.
 - ▶ You can choose from simple linear regression or multiple linear regression.
 - ▶ Compare your implementation with **scikit-learn** version and show that it works correctly.

Task description (2)

- ▶ Feature selection/engineering
 - ▶ Choose one variable (or more) which you want to predict using linear regression models.
 - ▶ Perform any preprocessing if necessary - remove outliers, fix missing data, etc...
 - ▶ Perform feature selection/engineering: various feature combinations, polynomial features, one-hot encoding, etc...
 - ▶ See if any findings from 1st task (EDA) could be applied to help select suitable features.
- ▶ Model evaluation & selection
 - ▶ Train linear regression models using different feature sets (**at least 3**).
 - ▶ Evaluate model performance: use regression metrics of your choice, visualize results, etc.
 - ▶ Compare performance of different models by using holdout or k-fold cross validation.

Task description (3)

- ▶ Result interpretation
 - ▶ 5 most important conclusions in domain terms
 - ▶ Connection between the numbers and charts and your own knowledge
 - ▶ Any unexpected, surprising or otherwise interesting facts about the data? Can you prove/disprove any of your initial assumptions based on the results?

Other info

- ▶ Maximum grade for the task: **1.0**
- ▶ Deadline for task submission to Gitlab: **2024-03-01 16:00**
- ▶ Task submission process:
 - ▶ Jupyter notebook + dataset (.csv file) uploaded to Gitlab, according to the instructions presented in lecture 1
 - ▶ Task presented and explained during exercise class (Tuesdays/Fridays, 18:00 - 19:30)